The Ising perceptron at nonzero margin: certified bounds along the capacity curve

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Abstract

The storage capacity of the Ising perceptron at margin κ is predicted by a replica-symmetric variational problem. Huang proved that the predicted threshold $\alpha_\star(\kappa)$ is an upper bound at every margin, provided four numerical hypotheses on explicit low-dimensional problems hold at that margin. We verify the four hypotheses in ball arithmetic at the margins $\kappa = -0.45$, -0.05, 0.05, 0.0995 and 0.13, so the upper bound holds there without conditions; for instance at $\kappa = 0.05$ the capacity is at most $\alpha_\star(0.05)$, which lies in $[0.781073068, 0.781074776]$. Three of the hypotheses are verified on the whole strip $\kappa \in [-0.65, 0.19]$. Two identities make this possible: the replica-symmetric free energy is stationary in both order parameters at its fixed point, and the derivative of the fixed-point map reduces to a one-term integral with positive integrand. For every $\kappa < 0$ we also prove a capacity lower bound by embedding a symmetric perceptron. The matching lower bound at $\alpha_\star(\kappa)$ remains open.

Keywords: binary perceptron, storage capacity, replica-symmetric formula, computer-assisted proof, ball arithmetic.
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1 Introduction

Store $m = \alpha N$ random patterns in a linear classifier with binary weights: for i.i.d. standard Gaussian vectors $g_1, \dots, g_m$ in $\mathbb{R}^N$ and a margin $\kappa \in \mathbb{R}$, ask whether some $w \in \{-1, +1\}^N$ satisfies

$$\frac{\langle g_a, w \rangle}{\sqrt{N}} \;\geq\; \kappa \qquad \text{for all } a \leq m.$$

Krauth and Mézard [1] computed the replica-symmetric prediction for the largest such α and conjectured it sharp; at $\kappa = 0$ their constant is $\alpha_\star = 0.8330786\dots$, in the margin framework of Gardner and Derrida [3]. Rigorous bounds at zero margin go back to Kim and Roche [4] and Talagrand [5]. Three decades later the zero-margin conjecture was resolved: Ding and Sun [6] proved the lower bound and Huang [7] the upper bound. Both proofs are conditional on numerical hypotheses about explicit low-dimensional variational problems, and the recent interval-arithmetic verification of those hypotheses [9] is a zero-margin verification: the certified statement stops at $\kappa = 0$.

Huang's upper-bound theorem, however, is stated at every margin. For each κ it bounds the capacity by $\alpha_\star(\kappa)$, the root of the replica-symmetric free energy along its fixed-point branch, conditionally on four numerical hypotheses at that margin. The first asks that the fixed-point equations have a unique solution, along which the free energy changes sign. The second asks that the associated approximate-message-passing iteration be stable at that solution. The third asks that an explicit function of one real variable, which controls the concavity of the first-moment functional near its maximizer, take a negative value at some point. The fourth asks that a rate function $\mathcal{S}_\star$ of two real variables be nonpositive on the whole plane (Conditions 1–4 of Section 2). At $\kappa = 0$ these are exactly the hypotheses closed in [9]; at $\kappa \neq 0$ they were open. Stojnic's fully-lifted random duality theory [17] predicts the same replica-symmetric values at every margin by nonrigorous means; the rigorous certification of the hypotheses had reached only $\kappa = 0$. We verify all four at five nonzero margins.

Main Theorem (Theorem 4)**.** *For $\kappa \in \{-0.45,\ -0.05,\ 0.05,\ 0.0995,\ 0.13\}$ the four hypotheses of Huang's theorem hold, and the threshold $\alpha_\star(\kappa)$ lies in the interval*

$$\begin{aligned} &\kappa = -0.45: \ [1.557652103,\ 1.557654133], \quad \kappa = -0.05: \ [0.889408031,\ 0.889409783],\\ &\kappa = 0.05: \ [0.781073068,\ 0.781074776], \quad \kappa = 0.0995: \ [0.733478513,\ 0.733480205],\\ &\kappa = 0.13: \ [0.705932217,\ 0.705933898]. \end{aligned}$$

For α above that interval, with probability $1 - o(1)$ as $N \to \infty$ no $w \in \{-1, +1\}^N$ satisfies all αN constraints.

This paper is a certified study of the whole margin axis. Its contributions are of three kinds.

First, two identities (Section 2.1) that make the general-margin system as well behaved as the zero-margin one. The replica-symmetric functional G is stationary in both order parameters exactly on the fixed-point branch, at every (α, κ); by the envelope theorem, along a differentiable fixed-point branch, $G'_\star(\alpha) = \mathbb{E} \log \Psi(u) < 0$ strictly, so on every certified rectangle (where the contraction makes the branch exist, stay unique, and vary smoothly) $\alpha_\star(\kappa)$ is the unique root of a strictly decreasing function. And the derivative $\partial R/\partial q$ of the constraint-side fixed-point map collapses, after an integration by parts, to a one-term integral with positive integrand. This is the general-margin form of Ding–Sun's monotonicity lemma, and the identity that makes a tight interval enclosure of the contraction possible at all.

Second, ball-arithmetic certificates (Sections 4 and 5.1) discharging the fixed-point, approximate message passing (AMP), and local-concavity hypotheses on the strip $\kappa \in [-0.65, 0.19]$, covered by 1680 contiguous slabs of width $5 \cdot 10^{-4}$. Uniqueness of the fixed point is carried by a per-slab contraction bound up to $\kappa = 0.10$ and, beyond the reach of that rectangle bound, by Nakajima's saddle-uniqueness theorem [8] under a premise certified in ball arithmetic. At five margins spanning the strip a certified bisection locates $\alpha_\star(\kappa)$ to width about $2 \cdot 10^{-6}$, and at six margins the Hessian of the first-moment functional at its degenerate maximizer is certified negative definite; this is the local ingredient of the global hypothesis.

Third, a certified verification of the global two-variable hypothesis at the five margins $\kappa = -0.45,\ -0.05,\ 0.05,\ 0.0995,\ 0.13$ (Sections 5.2 and 6). The zero-margin certificate architecture of [9], a compact moment-coordinate sweep plus ray concavity on a star around the degenerate maximizer, does not transfer as-is, because at $\kappa \neq 0$ the fixed-point parameters (q_0, ψ_0) are known only as located boxes of width up to 10^{-4}, four orders wider than the zero-margin rectangle, and that width drowns every value-level enclosure the original design relies on. The

port that works keeps the maximizer symbolic through the base-point identity $\nabla\Phi(1,0) = (\psi_0(1-q_0),\, q_0)$, evaluates the dual localization through a correlated Bregman integrand that collapses algebraically to $\log(\cosh\Delta + M\sinh\Delta) - M\Delta$, and stabilizes the far-tail second-derivative kernels by classical Mills-ratio bounds ($E' \in (0,1)$, $E'' \in (0, E-V)$). With those in place all four pieces of the zero-margin architecture certify at each of the five margins, and Huang's theorem yields unconditional upper bounds there.

The certified theorems are upper bounds. The matching lower bound at $\kappa \neq 0$ is open: the embedded bound of Section 8 holds for every $\kappa < 0$ but lies far below $\alpha_\star(\kappa)$, so the exact threshold at $\kappa \neq 0$ is not determined (Section 11).

Relation to the zero-margin verification

The certified primitives (ball-arithmetic special functions, rigorous quadrature, exact serialization) are those of [9], and at $\kappa = 0$ every pipeline in this paper reproduces the certified zero-margin values (the fixed-point rectangle, Huang's AMP and local-concavity constants, and his Hessian intervals), which we use throughout as anchors. The mathematical content specific to this paper is the pair of identities of Section 2.1, the Hessian formulas at general margin in Section 5.1, and the stability layer of Section 6; the engineering content is the located-box certificate architecture, which is what a margin interval demands once no analytically certified parameter rectangle exists.

2 The system and the four hypotheses

Let φ and Ψ denote the standard Gaussian density and upper tail, $E = \varphi/\Psi$ the inverse Mills ratio (an increasing function; $E' = E\,(E-u) \in (0,1)$, proved with its second-derivative bounds in Lemma 3), and $Z \sim N(0,1)$. Fix $\kappa \in \mathbb{R}$ and write

$$u(z; q, \kappa) = \frac{\kappa - \sqrt{q}\, z}{\sqrt{1-q}}, \qquad 0 < q < 1.$$

Following Ding–Sun and Huang, the replica-symmetric system is

$$P(\psi) = \mathbb{E}\big[\tanh^2(\sqrt{\psi}\, Z)\big], \qquad R(q,\alpha) = \frac{\alpha}{1-q}\,\mathbb{E}\big[E(u(Z))^2\big], \tag{1}$$

$$G(\alpha, q, \psi) = -\frac{\psi(1-q)}{2} + \mathbb{E}\log\big(2\cosh(\sqrt{\psi}\, Z)\big) + \alpha\,\mathbb{E}\log\Psi\big(u(Z)\big), \tag{2}$$

with the fixed-point branch $q = P(\psi)$, $\psi = R(q,\alpha)$ and the threshold $\alpha_\star(\kappa)$ defined as the zero of $G_\star(\alpha) = G(\alpha, q_\star(\alpha), \psi_\star(\alpha))$. At $\kappa = 0$, where $u = -\gamma z$ in law with $\gamma = \sqrt{q/(1-q)}$, this is the certified zero-margin system. (Huang's displayed free energy carries a sign typo on the α term; the fixed point requires the plus sign, which is what Ding–Sun use and what reproduces the certified rectangle.)

Huang's theorem bounds the capacity at margin κ by $\alpha_\star(\kappa)$ subject to four numerical hypotheses, which we state in the form his proof consumes.

Condition 1 (fixed point)**.** *On a stated rectangle, $q \mapsto P(R(q,\alpha))$ has a unique fixed point for every α in a stated interval, with $\sup_{q\in(q_{\mathrm{lb}},q_{\mathrm{ub}})}(P\circ R_\alpha)'(q) < 1$ uniformly on the rectangle, and $G_\star$ crosses zero inside that interval. The rectangle is existentially quantified: Huang's own $\kappa = 0$ instantiation (his Proposition 3.2) takes an interval of width 10^{-9} about the fixed point.*

Condition 2 (AMP). $\alpha\,\mathbb{E}[\mathrm{sech}^4(\sqrt{\psi_0}Z)]\;\mathbb{E}[E'(u(Z;q_0,\kappa))^2]/(1-q_0)^2 \;<\; 1$ *at the fixed point, with* $u(z;q,\kappa)=(\kappa-\sqrt{q}\,z)/\sqrt{1-q}$ *as above.*

Condition 3 (local concavity). $\ell(z) < 0$ *for some* $z > -1$*, where* ℓ *is the explicit function recalled in Section 4; a single witness suffices.*

Condition 4 (two-variable function). $\mathcal{S}_\star(\lambda_1,\lambda_2) \le 0$ *for all* $(\lambda_1,\lambda_2)\in\mathbb{R}^2$*, where* $\mathcal{S}_\star$ *is the rate function recalled in Section 5.2; equality holds at* $(1,0)$.

2.1 Two identities

The function G is that of Ding and Sun [6, eq. (4)], who define it at general margin; the two lemmas below record its partial derivatives and the derivative of R at every κ.

Lemma 1 (stationarity). *For all* $\alpha>0$*,* $\kappa\in\mathbb{R}$*,* $\psi>0$ *and* $0<q<1$*,*

$$\frac{\partial G}{\partial \psi}=\frac{q-P(\psi)}{2},\qquad \frac{\partial G}{\partial q}=\frac{\psi-R(q,\alpha)}{2}.$$

In particular the critical points of G are exactly the solutions of Ding and Sun's fixed-point system $q=P(\psi)$, $\psi=R(q,\alpha)$ [6], at every margin.

Proof of Lemma 1. The first identity is Gaussian integration by parts on the $\log\cosh$ term. For the second, differentiating u in q gives

$$u_q=\frac{1}{2\sqrt{1-q}}\Big(-\frac{z}{\sqrt{q}}+\frac{u}{\sqrt{1-q}}\Big),\qquad \frac{\partial}{\partial q}\,\mathbb{E}\log\Psi(u)=-\mathbb{E}\big[E(u)\,u_q\big].$$

Stein's lemma applied to the z part, $\mathbb{E}[zE(u)]=-\sqrt{q/(1-q)}\;\mathbb{E}[E'(u)]$, together with the Mills identity $E'(u)=E(u)^2-uE(u)$, collapses the two pieces:

$$\mathbb{E}[E(u)\,u_q]=\frac{\mathbb{E}[E'(u)]+\mathbb{E}[u\,E(u)]}{2(1-q)}=\frac{\mathbb{E}[E(u)^2]}{2(1-q)}=\frac{R(q,\alpha)}{2\alpha}.$$

Hence $\partial_q G=\psi/2-\alpha\cdot R/(2\alpha)=(\psi-R)/2$. □

Both derivatives vanish exactly on the fixed-point branch. On a rectangle carrying the contraction of Condition 1 the branch $\alpha\mapsto(q_\star,\psi_\star)$ is C^1 (the implicit function theorem applies: $1-(P\circ R_\alpha)'$ is bounded away from zero), and the envelope theorem gives $G'_\star(\alpha)=\mathbb{E}\log\Psi(u)<0$ strictly ($\Psi<1$ pointwise), for every κ: the root $\alpha_\star(\kappa)$ is unique on the branch. This is the general-margin form of the monotonicity clause in Condition 1, and it is also what turns the located fixed-point boxes of Section 4 into certified $\alpha_\star$ intervals: a certified sign of G at a located box is a certified side of $\alpha_\star$.

Lemma 2 (derivative of R). *For all* $\alpha>0$*,* $\kappa\in\mathbb{R}$ *and* $0<q<1$*,*

$$\frac{\partial R}{\partial q}=\frac{\alpha}{(1-q)^2}\,\mathbb{E}\Big[E'(u)^2+2\,E(u)^2E'(u)\Big]\;>\;0.$$

Proof. Differentiating (1) in q produces $R/(1-q)+\frac{\alpha}{1-q}\mathbb{E}[2EE'u_q]$. Splitting u_q as in Lemma 1 and applying Stein's lemma to $\mathbb{E}[z\,EE'(u)]$ (with $(EE')'=E'^2+EE''$ and $E''=E'(E-u)+EE'-E$) turns the cross term into

$$\mathbb{E}[2EE'u_q]=\frac{\mathbb{E}[E'^2+2E^2E'-E^2]}{1-q},$$

and the $-\alpha\mathbb{E}[E^2]/(1-q)^2$ piece cancels $R/(1-q)$ exactly. Positivity is immediate since $E'\in(0,1)$. In both proofs differentiation under the integral sign is dominated ($|z\tanh(\sqrt{\psi}z)|\le\sqrt{\psi}\,z^2$, and the q-integrands grow at most quadratically in $|z|$, locally uniformly in the parameters), and Stein's lemma applies because $E(u(z))$ and $(EE')(u(z))$ are absolutely continuous with bounded derivative in z. □

Lemma 2 replaces Ding–Sun's Lemma 7.2 (monotonicity of R, stated at $\kappa=0$) at every margin. The same expectation $\mathbb{E}[E'^2+2E^2E']$ appears in Nakajima's analysis of the replica-symmetric saddle [8, eq. (18)]. Its one-term form matters twice over: the integrand is a product of quantities with classical bounds, so its interval enclosure is tight enough for the per-slab contraction certificates; and positivity makes $q\mapsto P(R(q))$ increasing, which is what the contraction-free location of Section 4 bisects on. Both identities were checked against high-precision finite differences (agreement at the 10^{-15} level along the whole continuation) and symbolically.

3 The capacity curve and the condition map

Before certifying anything we map the terrain in high-precision (nonrigorous) arithmetic: a warm-started continuation of the fixed point along $\kappa\in[-2.0,1.5]$, recording at each of 71 points the threshold, the fixed point, the contraction $(P\circ R_\alpha)'(q_\star)$, the AMP product, the local-concavity minimum ℓ_0, and the stationarity residuals (which sit at the finite-difference noise floor, confirming Lemma 1 pointwise). Selected values:

κ	$\alpha_\star$	$q_\star$	$(P\circ R)'$	AMP	ℓ_0
-2.0	29.9921	0.11863	0.387	0.440	-0.147
-1.0	3.79463	0.36519	0.583	0.607	-0.099
-0.5	1.67884	0.47377	0.666	0.589	-0.132
0	0.83308	0.56395	0.744	0.544	-0.191
0.5	0.45566	0.63841	0.810	0.489	-0.271
1.0	0.27070	0.69988	0.863	0.433	-0.359
1.5	0.17245	0.75049	0.903	0.381	-0.442

All three per-point conditions hold at every point computed. The contraction is the only quantity trending toward failure, and only slowly; ℓ_0 approaches zero nearest $\kappa\approx-1.3$ (about -0.094) and then strengthens again. On the positive branch $\alpha_\star(\kappa)$ stays below Nakajima's bound at every grid point (worst ratio 0.874 at $\kappa=1.5$). Toward $\kappa=-2$, the lower edge of the computed range, the threshold approaches the annealed bound from below, as it must where the constraints trivialize. The optimal tilt of the two-variable functional at its maximizer matches $\sqrt{1-q_0}$ to eight digits at every κ checked, confirming numerically Huang's identity for the tilt (stated by him at general margin) along the curve; the identity is used exactly in Section 6.

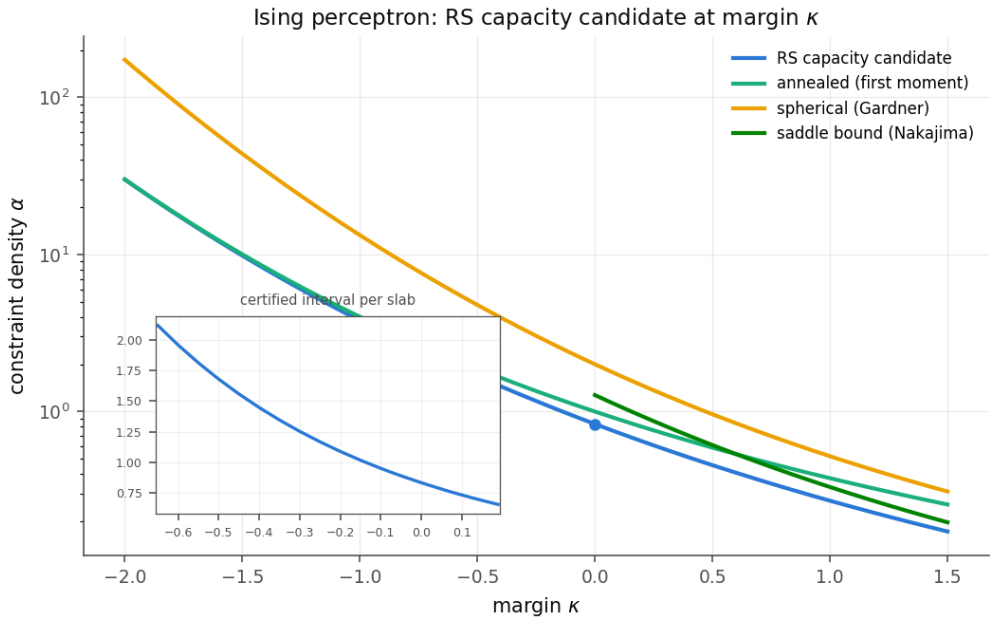


Figure 1: The replica-symmetric threshold $\alpha_\star(\kappa)$ (warm-started continuation from the certified value at $\kappa = 0$), the annealed bound $\log 2/(-\log \Psi(\kappa))$, the Gardner spherical capacity [2], and Nakajima's saddle-existence bound $\alpha_c(\kappa) = 2/(\pi\, \mathbb{E}[(\kappa - Z)_+^2])$ for $\kappa \geq 0$. The inset shows the certified per-slab $\alpha_\star$ intervals on the strip.

4 Certificates on a strip

For κ in sub-slabs of width $5 \cdot 10^{-4}$ covering $[-0.65, 0.19]$, with per-slab parameter rectangles seeded by the continuation (the certificate is the proof; a too-tight seed simply fails and is regenerated wider), we verify Conditions 1–3 in ball arithmetic (`python-flint`, 80-bit working precision, every input a ball covering the whole slab).

4.1 Numerical architecture

Three implementation laws carry the whole computation; each was learned from a measured failure, and the failed artifacts are kept in the repository as evidence.

First, adaptive complex quadrature cannot resolve enclosure width that comes from parameter balls: subdividing the integration variable does not shrink it, and the integrator exhausts any budget. So every wide-parameter integral is a real-cell Riemann sum with exact per-cell Gaussian masses. Second, the ball evaluation of φ/Ψ fails on wide arguments in the steep region (the enclosure of Ψ crosses zero), while the ratio is monotone; all Mills-family factors are therefore enclosed by two thin endpoint evaluations, and the sign-critical sums use the mean-value cell rule (midpoint value plus a derivative correction; on a cell of width h its error is $O(h^2)$ against the plain hull's $O(h)$, with constants set by the derivative enclosures on that cell). Third, expectations that nearly cancel must be evaluated as one correlated integrand, never as separate sums whose parameter widths add. For Condition 3 this is the identity (with

$\tau = E'$, $A = 1 - q_0$, and $b = \mu(\hat{z}) - A$ for $\mu(\hat{z}) = \mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0}Z))^{-1}]$)

$$\ell(\hat{z}) = \hat{z} + \alpha\,\mathbb{E}\left[\frac{b\,\tau^2}{A(A + b\tau)}\right],$$

whose integrand is monotone in τ for fixed (μ, A), regular on all of $\tau \in [0, 1]$, and bounded by $|b|/(A\mu)$; the separated form's enclosure crossed zero on the $[-0.45, -0.35]$ batch while the true value stays near -0.14. The identity, the monotonicity, and the bound are verified symbolically.

4.2 The contraction lane

On each slab the checks are: (i) the contraction $\sup_q (P \circ R_\alpha)' \le \sup_\psi P' \cdot \sup_q |\partial R/\partial q| < 1$ on the slab rectangle, via Lemma 2; (ii) existence: the endpoints of the q interval map strictly inward under $P \circ R_\alpha$ on each of eight α sub-intervals; (iii) the crossing: at the two thin α endpoints a certified Picard iteration locates the fixed point (the invariant is that the image of any set containing the fixed point contains it) and the sign of G resolves through a mean-value form that keeps the stationarity cancellations of Lemma 1 explicit, $G(\alpha_{\mathrm{lb}}) > 0 > G(\alpha_{\mathrm{ub}})$; (iv) Condition 2; (v) Condition 3 at Huang's witness $\hat{z} = -0.6693$. Together with the strict decrease of $G_\star$, (i)–(iii) discharge Condition 1 verbatim on the slab: $\alpha_\star(\kappa)$ exists, is unique, and lies in the slab's α interval for every κ in the slab.

The contraction bound is claimed per slab only. Over a batch hull the κ width poisons the $\partial R/\partial q$ enclosure and the product of suprema loses a factor of about two against the supremum of the product; on a single slab the identity-based enclosure clears 1 with margin 0.049 at $\kappa = 0$, growing to 0.198 at $\kappa = -0.65$ and thinning to 0.0083 at the positive edge of the lane.

All 1500 slabs covering $[-0.65, 0.10]$ pass every check, in forty independent worker runs whose collector also gates contiguity (consecutive slabs must share their κ boundary). Two first attempts failed and are kept as evidence: one at doubled slab width (the q pad must cover the fixed point's drift across the slab's (α, κ) rectangle, dominated by $\partial q_\star/\partial\alpha \approx 0.92$ times the α pad; the direct drift $dq_\star/d\kappa$ stays near 0.18 across the strip), and one on the separated-form ℓ enclosure described above.

4.3 The Nakajima lane

The rectangle product bound saturates near $\kappa \approx 0.12$, and we measure that this wall belongs to the method, not the condition: at $\kappa = 0.13$ the box product reads 1.32, while a certified sign bisection of the displacement $P(R(q)) - q$ (which has certified signs at the interval endpoints and, by the uniqueness of the fixed point, a single zero between them, so it is positive below the fixed point and negative above) locates the fixed point at both α endpoints to width $2 \cdot 10^{-8}$ with no contraction input, the contraction evaluated on the located intervals reads 0.79, and the G crossing resolves with margins $2 \cdot 10^{-3}$.

The rectangle clause itself is then discharged at $\kappa = 0.13$ exactly as Huang's Proposition 3.2 discharges it at $\kappa = 0$: on the rectangle $\alpha \in [0.705932217, 0.705933898]$, $q \in [0.5836943, 0.5857104]$ (the located fixed points $\pm 10^{-3}$), the certified contraction product is 0.8435, the map is endpoint-inward at both α ends with margins above $2.3 \cdot 10^{-4}$, the G signs at the located fixed points are $+1.6 \cdot 10^{-7}$ and $-1.6 \cdot 10^{-6}$, and $\mathbb{E} \log \Psi < 0$ over the rectangle (`rect_cond31_0p13.json`; locate and G evaluator depth 16000). The wide-rectangle product

1.32 is a pad artifact, not a property of the map, and the map itself carries no wall across the lane: the true fixed-point contraction $(P \circ R_\alpha)'(q_\star)$ is nearly flat, rising only from 0.758 at $\kappa = 0.10$ to 0.770 at $\kappa = 0.19$, everywhere below 1 with margin above 0.22. What decides the rectangle clause is therefore the pad width, not the margin: a probe over the lane (`rect_strip_nak.jsonl`, per κ sub-ball) clears 1 at the lower edge with a $2 \cdot 10^{-3}$ half-pad but crosses it (1.012) by $\kappa = 0.15$ as that same pad inflates the flat 0.765, exactly as a thinner located rectangle pulls $\kappa = 0.13$ back to 0.844. The pad that matters is the located q interval, whose width is set mostly by the slab's α width ($3 \cdot 10^{-3}$) through $\partial q_\star / \partial \alpha \approx 0.9$; narrowing it means deep-locating $\alpha_\star$ per slab to $\sim 10^{-6}$, the same tightening the five point margins already carry, after which the flat contraction leaves a thin rectangle that clears 1. We first ran this recipe at both ends of the lane and an interior point ($\kappa \in \{0.13, 0.15, 0.185\}$; contractions 0.844, 0.836, 0.830 on deep-located rectangles; `rect_cond31_0p15.json`, `rect_cond31_0p185.json`), and then per slab across the whole lane: on each of the 180 slabs, over eight κ sub-balls of width $6.25 \cdot 10^{-5}$, the located rectangle with half-pad $5 \cdot 10^{-4}$ is endpoint-inward at both α ends and carries a certified contraction product below one, the worst bound across the lane being 0.958 (`rect_strip_nak_thin_*.jsonl`, collected with the input hashes bound in `rect_strip_thin_summary.json`). With the inward endpoints and Lemma 2's monotonicity this gives, for every κ in the lane, a rectangle on which the fixed point is unique by the contraction itself, so the rectangle clause of Condition 1 holds on every slab with uniqueness carried both by the rectangle and, independently, by Nakajima's theorem.

The second lane certifies $[0.10, 0.19]$ (180 slabs) on that design: existence by the same inward sweep; location by certified sign bisection per κ sub-ball (the κ width floors the displacement's sign resolution, so each slab runs eight sub-balls whose union covers it); the contraction consumed only on located intervals, which is the local form Huang's perturbation argument uses; and uniqueness from Nakajima's saddle theorem [8], whose premise $\alpha_{\mathrm{ub}} < \alpha_c(\kappa) = 2/(\pi\, \mathbb{E}[(\kappa - Z)_+^2])$ is certified per slab through the closed form $\mathbb{E}[(\kappa - Z)_+^2] = (1+\kappa^2)\big(1 - \Psi(\kappa)\big) + \kappa\varphi(\kappa)$ (worst certified premise margin 0.311).

Across both lanes: 1680 contiguous slabs, zero failures. At five margins a certified bisection on the sign of G at located boxes tightens the slab intervals to width about $2 \cdot 10^{-6}$:

$\alpha_\star(0.05) \in [0.781073068,\ 0.781074776]$, $\alpha_\star(-0.05) \in [0.889408032,\ 0.889409783]$,
$\alpha_\star(0.0995) \in [0.733478514,\ 0.733480204]$, $\alpha_\star(-0.45) \in [1.557652103,\ 1.557654133]$,
$\alpha_\star(0.13) \in [0.705932218,\ 0.705933898]$.

Each interval contains its continuation value; the slab containing $\kappa = 0$ contains the Ding–Sun constant. The bisection floor is measured, not guessed: at the default quadrature depth the G enclosure binds at width $1.4 \cdot 10^{-5}$, and raising the two quadrature floors breaks through exactly as budgeted.

5 The two-variable function

5.1 The maximizer and its Hessian

The local ingredient of Condition 4 is the negative definiteness of the Hessian $\mathcal{H}$ of Huang's first-moment functional at its degenerate maximizer. His second-derivative expansion, *before* the zero-margin specialization, gives at every κ the same quadratic form

$$\langle w, \mathcal{H} w \rangle = -(I_1 w_1^2 + 2 I_2 w_1 w_2 + I_3 w_2^2) + C_1 a^2 + C_2 ab + C_3 b^2,$$

where $a = I_1 w_1 + I_2 w_2$ and $b = I_2 w_1 + I_3 w_2$, with prior-side integrals I_j unchanged in form ($I_1 = \psi_0(1-q_0) - 2\psi_0^2(1-4q_0+3p_4)$, $I_2 = \psi_0(1-4q_0+3p_4)$, $I_3 = q_0 - p_4$, $p_4 = \mathbb{E}\tanh^4(\sqrt{\psi_0}Z)$) and constraint-side constants C_j that at $\kappa \neq 0$ must be taken from the pre-specialization display: his zero-margin closed forms rest on two identities of the unshifted law, of which $\alpha\,\mathbb{E}[\eta^2] = \psi_0$ survives at every margin (it is the same identity behind the unbounded-tilt directions of Section 5.2) while $\alpha\,\mathbb{E}[\eta\hat{H}] = -q_0(1-q_0)\psi_0$ does not. We evaluate the general-margin C_j directly at the shifted law. With $\hat{H} = \sqrt{q_0}\,Z$, $u = (\kappa - \hat{H})/\sqrt{1-q_0}$, $\eta = E(u)/\sqrt{1-q_0}$, $F' = -E'(u)/(1-q_0)$, and $Q = -\hat{H}/q_0 + \big(\kappa - \hat{H} - (1-q_0)\eta\big)/(1-q_0)$, every expectation over the single Gaussian $\hat{H}$,

$$C_1 = \frac{\alpha}{\psi_0^2}\,\mathbb{E}[F'\eta^2], \qquad C_2 = -\frac{2\alpha}{\psi_0}\,\mathbb{E}[F'NQ] + \frac{2}{1-q_0},$$

$$C_3 = \alpha\,\mathbb{E}[F'Q^2] + \frac{2\alpha}{q_0(1-q_0)}\,\mathbb{E}[\eta\hat{H}] - \alpha\Big(\frac{3}{(1-q_0)^2} + \frac{1}{q_0(1-q_0)}\Big)\mathbb{E}\big[\eta\,(\kappa - \hat{H} - (1-q_0)\eta)\big].$$

The certified evaluation encloses each expectation by the mean-value cell rule with explicit quadratic tail majorants (η grows at most linearly on the left tail, F' is bounded by $1/(1-q_0)$ in magnitude, and every integrand is degree at most two in $|z|$), and the near-cancelling prior moments are evaluated as single correlated integrands. The certified enclosures at the six margins are one-sided where sign is the claim:

$$\begin{array}{lll}
\kappa = -0.45: & \mathcal{H}_{11} \le -0.0647, & \det\mathcal{H} \ge 2.9\cdot 10^{-4},\\
\kappa = -0.05: & \mathcal{H}_{11} \le -0.0471, & \det\mathcal{H} \ge 1.6\cdot 10^{-4},\\
\kappa = 0: & \mathcal{H}_{11} \le -0.0451, & \det\mathcal{H} \ge 1.8\cdot 10^{-4},\\
\kappa = 0.05: & \mathcal{H}_{11} \le -0.0424, & \det\mathcal{H} \ge 1.2\cdot 10^{-4},\\
\kappa = 0.0995: & \mathcal{H}_{11} \le -0.0400, & \det\mathcal{H} \ge 1.0\cdot 10^{-4},\\
\kappa = 0.13: & \mathcal{H}_{11} \le -0.0385, & \det\mathcal{H} \ge 8.8\cdot 10^{-5},
\end{array}$$

the trend, both margins thinning as κ grows, being the quantitative form of the flattening of F near its maximizer that Section 7.1 prices.

Certification requires parameter boxes far tighter than the slabs provide ($\det\mathcal{H} \sim 10^{-4}$ moves at order $dC/dq \sim 10$), which is what the deep $\alpha_\star$ intervals above are for. On the resulting located boxes ($3.6\cdot 10^{-6}$ in q_0, $5.2\cdot 10^{-5}$ in ψ_0 at $\kappa = 0.05$) the certificate resolves: $\mathcal{H}_{11} < 0$ and $\det\mathcal{H} > 0$ in ball arithmetic, with the near-cancellations $1 - 4q_0 + 3p_4$ and $q_0 - p_4$ evaluated as single correlated integrands and the five constraint-side integrals by mean-value Riemann sums with elementary Mills-family derivatives. The same chain certifies at $\kappa = 0$ on the Ding–Sun rectangle, where the C_j enclosures land inside Huang's certified intervals (an independent re-verification of his claim through the general-margin formulas), and at -0.05, 0.0995, -0.45 (where $C_1 = -0.855$ against -0.718 at zero margin: genuinely shifted territory, not a perturbation), and 0.13 past the product-bound wall. Six margins in all.

Along the continuation the fixed-tilt $\mathcal{H}$ stays negative definite up to $\kappa = 1.15$ and loses definiteness inside $(1.1812, 1.1844)$, while the Hessian of the tilt-optimized functional remains negative out to at least $\kappa = 1.5$; the two are consistent by the Schur relation (the fixed-tilt form dominates), so past $\kappa \approx 1.18$ a local argument must re-optimize the tilt, as the ray certificates below already do.

5.2 The function at general margin

Huang's rate function at margin κ, in the form his proof consumes, is

$$\mathcal{S}(\Lambda, s) = \frac{s^2\psi_0}{2} + \mathrm{ent}(\Lambda) + \alpha\,\mathbb{E}\log\Psi\left(\frac{\kappa - (a_2/q_0)\hat{H} - (a_1/\psi_0)\eta}{D} + s\eta\right),$$

where $\dot{H} \sim N(0,\psi_0)$ and $\hat{H} \sim N(0,q_0)$ are independent, $M = \tanh\dot{H}$ (Huang's notation), $\eta = F_{1-q_0}(\hat{H}) = E\big((\kappa-\hat{H})/\sqrt{1-q_0}\big)/\sqrt{1-q_0}$ is the shifted cavity law (this is one of the two places the margin enters; the other is the additive κ in the $\log\Psi$ argument), $a_1 = \mathbb{E}[\dot{H}\Lambda]$, $a_2 = \mathbb{E}[M\Lambda]$, $D = \sqrt{1-a_2^2/q_0}$, and $\mathcal{S}_\star(\lambda_1,\lambda_2) = \inf_{s\geq 0}\mathcal{S}(\tanh(\lambda_1\dot{H} + \lambda_2 M), s)$. Condition 4 asks $\mathcal{S}_\star \leq 0$ everywhere.

The zero-margin verification's reduction applies verbatim: the profile enters only through the moment pair (a_1, a_2), which ranges over a compact convex body K (the moment body of $(\dot{H}, M)$), and on K the entropy term is bounded by convex duality, $H(a) \leq \Phi(b) - b\cdot a$ for every dual point b, with $\Phi(b) = \mathbb{E}\log 2\cosh(b_1\dot{H} + b_2 M)$ free of κ. The margin lives entirely in the constraint term. Two structural facts organize the verification. The maximizer sits at $a^\star = (\psi_0(1-q_0),\, q_0)$ for every κ: this is the base-point identity $\nabla\Phi(1,0) = (\mathbb{E}[\dot{H}M], \mathbb{E}[M^2]) = (\psi_0(1-q_0), q_0)$ (Gaussian integration by parts and the fixed-point equations), and Huang's lemma gives $\mathcal{S}_\star = 0$ there with optimal tilt $s_0 = \sqrt{1-q_0}$, both in closed form. And on roughly the half-plane $\lambda_1 < 0$ the s-infimum is genuinely $-\infty$: $\alpha\,\mathbb{E}[\eta^2] = \psi_0$ exactly at the fixed point, so the quadratic tilt coefficient cancels, and wherever the surviving linear coefficient of the tilt is nonpositive the functional runs away (the measured boundary of that region at $\kappa = 0.05$ crosses $\lambda_1 = 0.04$ on the $\lambda_2 = 0$ axis and bends left as $|\lambda_2|$ grows). Nothing downstream consumes this observation: the condition asks for the uncapped infimum over $s \geq 0$, so a certified negative value at one explicit finite tilt (which is what the sweep certificates exhibit cell by cell) is already a complete witness, and the runaway directions are the capped-tilt case handled inside Huang's own argument.

Float diagnostics along the curve (the certified analogue follows) show a function essentially independent of κ: a single peak at $(1,0)$, off-center values below $-3\cdot 10^{-3}$, negative far rays, Hessian eigenvalues near $(-0.22, -0.0045)$, and moment-body clearance near 0.017, with the weak eigenvalue degrading toward deep negative margin (-0.0013 at $\kappa = -1.5$), the measured approach to the expected replica-symmetry-breaking territory.

6 Certifying the two-variable condition

The zero-margin certificate has four pieces: a sweep of the moment body outside an exclusion box around $a^\star$ (Region II), banded ray-concavity certificates on a star around $a^\star$ (Region I), a second sweep of the exclusion box minus the star (stage 2), and an interior-ball certificate placing the star strictly inside the moment body. The obstruction at $\kappa \neq 0$ is uniform across all four: at zero margin the parameters (q_0, ψ_0) are certified to a rectangle of width 10^{-9}, so the maximizer's coordinates can be stored as decimals and every enclosure treats them as exact; at $\kappa = 0.05$ the located boxes have width up to $1.9\cdot 10^{-4}$ in ψ_0, and that width enters every plain evaluation of Φ at about $2\cdot 10^{-5}$, fatally wide next to valley margins of order 10^{-7} in the dual localization. Three devices repair the architecture; each is exact mathematics, not tuning.

The symbolic maximizer. Every certificate that references $a^\star$ or the base tilt s_0 consumes them as the parameter balls $(\,\psi_0(1-q_0),\, q_0\,)$ and $\sqrt{1-q_0}$, never as stored decimals, so the true maximizer, at the true parameters, is covered exactly and the ray identities ($\mathcal{S}=0$, $\nabla\mathcal{S}=0$, tilt optimality at the true $a^\star$) hold on the certified rays with no origin-inflation ball. Ray coverage is exact: certificates evaluated over the origin ball cover in particular the rays from the true origin.

The ray argument. For a unit direction v and a slope $c\in\mathbb{R}$ put $\chi_{v,c}(t)=\mathcal{F}(a^\star+tv,\, s_0+ct)$, where $\mathcal{F}(a,s)$ is the moment-coordinate form of $\mathcal{S}(\cdot,s)$ that the certificates evaluate, so that $\mathcal{S}_\star(a^\star+tv)\le\chi_{v,c}(t)$ whenever $s_0+ct\ge 0$. At the true parameters $\chi_{v,c}(0)=\chi'_{v,c}(0)=0$ for every c, by the value, gradient and tilt identities at $a^\star$. If $\chi''_{v,c}\le 0$ on $[0,T(v)]$ then $\chi_{v,c}\le 0$ there, hence $\mathcal{S}_\star\le 0$ on the segment. The chain needs one slope per ray: the second-derivative bound on a radial band $[t_0,t_1]$ with $t_0>0$ says nothing about χ itself unless χ and χ' are controlled at t_0, and those are controlled only through the same function on $[0,t_0]$. The certificates therefore fix the slope by direction alone: the circle is cut into 768 equal sectors, each sector carries one slope (the optimal tilt slope at its midpoint, rounded to six decimals, recorded in the manifest), every angular cell of every radial band lies inside one sector and uses that sector's slope, and a direction on a sector edge is covered by the cells of either neighbor, each with its own fixed slope. The checker recomputes each cell's sector from its angles, confirms that the cells of each band tile its angle range, and confirms the slope table is the same in the main and supplementary manifests.

The correlated Bregman evaluator. The dual localization (the argument that the dual maximizer $\lambda(x)$ stays inside a certified box, so that the entropy Hessian can be bounded there) needs signs of Bregman gaps $\Phi(\lambda)-\Phi(\hat\lambda_k)-(\lambda-\hat\lambda_k)\cdot x$ with $x=a^\star+\text{offset}$. Evaluated as differences of plain Φ enclosures these carry the full parameter fuzz twice. Written against the anchor $b_0=(1,0)$, the integrand collapses exactly: with $\Delta=(\lambda_1-1)\dot H+\lambda_2 M$,

$$\log\frac{\cosh(\dot H+\Delta)}{\cosh\dot H}-(\lambda-b_0)\cdot\nabla_b\log 2\cosh(b\cdot(\dot H,M))\big|_{b_0}=\log\big(\cosh\Delta+M\sinh\Delta\big)-M\Delta,$$

because $(\lambda-b_0)\cdot\nabla h(b_0)=M\Delta$ pointwise. Every factor is a small ball with thin coefficients, so the parameter width enters only at second order in $|\lambda-b_0|$; at the box corners, where plain differences of enclosures fail, the enclosures are two orders of magnitude tighter, and at the anchor the evaluator returns the exact zero ball. Bregman gaps between arbitrary points are then differences of two anchored gaps, and the gradient term the localization needs is the same trick applied to $\nabla\Phi$: $\nabla\Phi(\lambda)-a^\star$ has the exact per-node form $(\dot H,M)\,(1-M^2)\tanh\Delta/(1+M\tanh\Delta)$.

The factor $\cosh\Delta+M\sinh\Delta$ itself has two exact evaluations with disjoint stability regimes: the cosh/sinh form is exact at $\Delta=0$ (which the inner-band gaps of order 10^{-7} require) but subtracts two $e^{|\Delta|}$-sized balls when $M\approx\pm1$ and $|\Delta|$ is large, where the guard against the complex logarithm's branch then fires on every subdivision; the split $e^{\Delta}(1+M)/2+e^{-\Delta}(1-M)/2$ has both terms nonnegative on the real axis but carries first-order M-width near $\Delta=0$. The evaluator branches per node at $|\Delta|=1$; both forms are identities, so the branch affects tightness only.

Stable far-tail kernels. The ray certificates consume second derivatives of the constraint term, whose kernels involve $E'(V)=E(V)(E(V)-V)$ and $E''(V)$, with E the inverse Mills ratio of Section 2, evaluated at cavity arguments V that reach about 20 on far quadrature cells; there $E(V)-V$ is a difference of two large balls and the kernels explode (we measured

certified upper bounds of $+2 \cdot 10^3$ on cells whose true ray second derivative is -6.6). Three elementary bounds, applied as intersections with the computed balls, remove the failure mode entirely.

Lemma 3 (Mills clamps)**.** *Let E be the inverse Mills ratio and $\sigma = E - V$. For every $V \in \mathbb{R}$,*

$$\sigma > 0 \quad \Big(\textit{and } \sigma < \frac{1}{V} \textit{ for } V > 0\Big), \qquad E' = E\sigma \in (0,1), \qquad 0 < E'' < \sigma.$$

Proof. For $\sigma > 0$: when $V \le 0$ this follows from $E > 0$, and when $V > 0$ integrating $\varphi(z) \le (z/V)\varphi(z)$ over $z \ge V$ gives $\Psi(V) < \varphi(V)/V$. For $\sigma < 1/V$: the function $r(V) = \Psi(V) - V\varphi(V)/(V^2+1)$ has $r'(V) = -2\varphi(V)/(V^2+1)^2 < 0$ and vanishes at $+\infty$, so $r > 0$, which rearranges to $E < V + 1/V$. Next, $E' = E^2 - VE = E\sigma > 0$, while the conditional moments $\mathbb{E}[Z \mid Z > V] = E$ and $\mathbb{E}[Z^2 \mid Z > V] = 1 + VE$ give $1 - E' = \mathrm{Var}(Z \mid Z > V) > 0$. The upper bound on E'' is the identity $E'' - \sigma = (E'-1)(E+\sigma)$ with $E' < 1$ and $E + \sigma > 0$.

Positivity of E'', the convexity of the inverse Mills ratio, is the one clause that takes an argument. Differentiating E' gives $E'' = E\,[\,\sigma(E+\sigma) - 1\,]$, and for W distributed as $Z - V$ conditioned on $\{Z > V\}$ the moment formulas above read $\mathbb{E}W = \sigma$ and $\mathbb{E}[W^2] = 1 - VE + V^2$, whence $\sigma(E+\sigma) - 1 = 2(\mathbb{E}W)^2 - \mathbb{E}[W^2]$: the claim is that this residual law has coefficient of variation below one. Its survival function is $\mathbb{P}(W > w) = \Psi(V+w)/\Psi(V) = e^{-\Gamma(w)}$ with $\Gamma(w) = \int_0^w E(V+x)\,dx$ convex, since $E' > 0$. Set $\theta = 1/\sigma$, so the rate-θ exponential has the same mean, and $J(w) = \Gamma(w) - \theta w$. Then J is convex with $J(0) = 0$, $J'(0) = E - 1/\sigma < 0$ (this is $E\sigma = E' < 1$ again), and $J(w) \to \infty$, so $J < 0$ exactly on some interval $(0, w_0)$ and $J > 0$ past it. Hence $\mathbb{P}(W > w) - e^{-\theta w} = e^{-\theta w}(e^{-J(w)} - 1)$ is positive before w_0 and negative after, while $\int_0^\infty (\mathbb{P}(W > w) - e^{-\theta w})\,dw = \sigma - \sigma = 0$. Therefore

$$\tfrac{1}{2}\,\mathbb{E}[W^2] - \sigma^2 = \int_0^\infty w\,\big(\mathbb{P}(W > w) - e^{-\theta w}\big)\,dw = \int_0^\infty (w - w_0)\big(\mathbb{P}(W > w) - e^{-\theta w}\big)\,dw < 0,$$

the last integrand being nonpositive and not almost everywhere zero. So $\mathbb{E}[W^2] < 2(\mathbb{E}W)^2$, that is $E'' > 0$, at every real V. The algebraic steps (the E'' formula, the equivalence, and the moment identities) are verified symbolically (`sympy_identities.py`). □

Intersecting a computed enclosure with a proven bound is always sound; on benign cells the computed ball is returned unchanged, and the $\kappa = 0$ regression is digit-identical.

With these three devices the four pieces certify at each of the five margins. We give the counts at $\kappa = 0.05$; the other margins are in Section 7. At $\kappa = 0.05$, on the located boxes and the deep $\alpha_\star$ interval of Section 4:

(i) *Region II*: 1296 top cells over the moment rectangle minus the exclusion box, refining to 855 leaves, zero failures.

(ii) *Region I*: the banded ray-concavity certificates over the star (radius 0.012 along the weak direction), 1404 band jobs from the symbolic origin, zero failures; the thinnest certified band margin is $-7.5 \cdot 10^{-5}$ and the deepest $-1.6 \cdot 10^2$. A supplementary run of twelve shoulder chunks extends the certified reach to radius 0.0105 on the arcs 0.14 to 0.19 radians off the weak axes (margins $-3.5 \cdot 10^{-3}$ to -1.9): the zone policy, not the concavity, was the binding constraint there, and stage-2 leaves straddling the wedge shoulder need the extra reach.

(iii) *Stage 2*: the exclusion box minus the star, 1363 leaves, zero failures, with star containment read from the Region-I manifests themselves: a certified band covers every direction of its chunk out to the star boundary, so the certified reach is the pointwise union of the main zone reach and the supplementary arcs, evaluated exactly on a piecewise-constant splitting; the SHA-256 of both manifests is bound into the stage-2 artifact.

(iv) *Star interior*: the star, origin ball included, lies strictly inside the moment body: constructive inradius 0.01481 against required 0.01209.

Together: $\mathcal{S}_\star \le 0$ on the whole plane at $\kappa = 0.05$. All four of Huang's hypotheses hold there, and his theorem gives the unconditional bound.

Theorem 4. *The storage capacity of the Ising perceptron at margin κ is at most $\alpha_\star(\kappa)$ for every $\kappa \in \{-0.45, -0.05, 0.05, 0.0995, 0.13\}$, with the certified enclosures*

$$\begin{aligned}
&\alpha_\star(-0.45) \in [1.557652103,\ 1.557654133], \quad \alpha_\star(-0.05) \in [0.889408031,\ 0.889409783],\\
&\alpha_\star(0.05) \in [0.781073068,\ 0.781074776], \quad \alpha_\star(0.0995) \in [0.733478513,\ 0.733480205],\\
&\alpha_\star(0.13) \in [0.705932217,\ 0.705933898]:
\end{aligned}$$

for α above the corresponding interval, with probability $1 - o(1)$ no $w \in \{-1, +1\}^N$ satisfies all αN constraints.

The same pipeline is parameterized over the located boxes of any certified slab; what stands between the point theorem and a κ-interval version of Condition 4 is quantified in Section 7.1. The other three conditions do not wait for that extension: they are certified over the whole strip, which gives the conditional structure of the program a closed form.

Theorem 5. *For every $\kappa \in [-0.65, 0.10]$, Conditions 1–3 hold, with the uniform rectangle contraction of Condition 1 certified on the slab rectangle itself. For every $\kappa \in (0.10, 0.19]$, Condition 1 holds per slab in full (existence, crossing, and the rectangle contraction on a thin located rectangle over κ sub-balls, with uniqueness from the rectangle contraction and, independently, from Nakajima's theorem under a certified premise), and Conditions 2 and 3 hold per slab. In every case $\alpha_\star(\kappa)$ exists, is unique on the fixed-point branch, and lies in the certified α interval of the slab containing κ. Consequently, at every κ at which Condition 4 and the full Condition 1 both hold (in particular the five margins of Theorem 4), and for every α above that slab interval's upper endpoint, with probability $1 - o(1)$ no $w \in \{-1, +1\}^N$ satisfies all αN constraints at margin κ.*

Condition 4 is certified at the five margins of Theorem 4 and at $\kappa = 0$, where our pipeline independently reproduces the zero-margin verification; at those six values the hypothesis of Theorem 5 discharges and the two theorems agree. Everywhere else on the strip, the entire distance between the conditional and unconditional statements is one two-variable inequality at that single margin.

7 Further margins

The four conditions of Theorem 4 are margin-wise statements, and the certificate pipeline of Sections 5.2–6 runs unchanged at any κ inside the strip where the fixed point is certified

κ	$\alpha^*(\kappa)$ window	sweep	Region I	stage 2	inradius / required
0.05	[0.781073068, 0.781074776]	855	1404 (+12)	1363	0.01481 / 0.01209
−0.45	[1.557652103, 1.557654133]	655	1404 (+12)	1196	0.01480 / 0.01207
−0.05	[0.889408031, 0.889409783]	867	1404 (+12)	1282	0.01527 / 0.01209
0.0995	[0.733478513, 0.733480205]	904	1404 (+12)	1390	0.01459 / 0.01209
0.13	[0.705932217, 0.705933898]	874	1404 (+12)	1422	0.01446 / 0.01210

Table 1: Certified leaf and band counts per piece, with zero failures in every manifest. Region I counts marked (+12) include the twelve shoulder-supplement chunks; under the enlarged exclusion box every margin's stage-2 lattice consumes one. The final column gives the constructive moment-body inradius against the required star radius, both exact-arithmetic bounds.

unique: the pilot locates the (q_0, ψ_0) boxes at both endpoints of the α window, configures the exact-arithmetic modules from those balls, and runs the same four pieces. Besides $\kappa = 0.05$ it certifies Condition 4 at $\kappa = -0.45$, -0.05, 0.0995 and 0.13. At the two negative margins the fixed-point uniqueness input comes entirely from the contraction certificates of Section 4. Table 1 lists the certified pieces at the five margins.

Every margin's Region-I manifest carries per-band interval packets for the frozen-tilt matrix B, and the kernel-checked skeleton includes for each a fifth family: Loewner definiteness of every band's B-matrix, decided over the integers in thirteen chunks of 108 bands together with a decided total count (55 kernel theorems per margin; the mutation battery rejects a sign corruption in each family).

The stage-2 containment reads each supplement arc on both angular branches, so a cell sees every arc whichever branch its interval lies on. At $\kappa = 0.0995$ the exclusion box has upper offset $+0.1030$. Just above the smaller offset $+0.0961$ the dual certificate of Region II degenerates along a thin ridge: on a cell of width 0.0018 the certified value interval is $[\pm 0.181]$, and the bound is negative again two cell-widths higher. Enlarging the box needs no change to Region I, because the stage-2 containment derives its tiling from the box and fails if the enlarged region leaves what the star and the supplement cover.

7.1 From points to intervals

The natural strengthening of Theorem 4 replaces each margin by a κ-interval, so that $\alpha_\star(\kappa)$ becomes an unconditional bound on a continuum. For Conditions 1–3 this is what the strip certificates already provide: those inequalities hold with margins of order 10^{-1}, and enclosing every input over a κ-slab costs far less than that. Condition 4 is different in kind, and it is worth recording precisely why, because the obstruction is quantitative.

Certifying $S_\star \le 0$ over a slab means every certificate must hold with the located parameters (α, q_0, ψ_0) replaced by their hulls across the slab. Three measurements fix the scales. First, with the present locating algorithm and quadrature budget the hull widths do not shrink below a floor (the inward fixed-point test resolves no finer than the quadrature fuzz, about $3.5 \cdot 10^{-4}$ in q_0, and $\alpha_\star$ itself drifts at $|d\alpha_\star/d\kappa| \approx 1$–3.4 across any slab), which translate into a moment-plane origin ball of radius at least a few times 10^{-3} however narrow the slab. Second, the evaluation fuzz of every certificate grows linearly with that ball: rerunning the $\kappa = -0.65$ sweep with slab-hulled parameters (ball $2.3 \cdot 10^{-2}$) inflates cell enclosures to ± 1.3, and the scaling back to the floor ball leaves ± 0.2. Third, the certificates that matter are thin: the

sweep's weakest certified cells at point margins sit at $-5 \cdot 10^{-4}$, and Region I's thinnest bands at $-7.5 \cdot 10^{-5}$, because the two-variable function is genuinely flat at its degenerate maximizer. A 10^{-4}-margin certificate cannot survive 10^{-1}-scale fuzz, at any slab width, with this evaluator architecture.

The same comparison explains why the strip conditions posed no such problem and points at the correct construction. Near the maximizer the certified Hessian gives quadratic control whose constants vary slowly in κ (the fixed-tilt determinant moves by about $2 \cdot 10^{-4}$ per unit margin), so a κ-uniform neighborhood estimate is available from point certificates plus a crude κ-derivative bound; crude is enough, because a Lipschitz constant needs no thin margin. Far from the maximizer the certificates are fat and absorb hull fuzz directly. What remains is the intermediate annulus, where quadratic control must be extended beyond the star radius with explicit remainder bounds before the two regimes meet. This estimate is the open mathematical step of the interval problem: the obstruction is measured and the two outer regimes are in hand, but a remainder bound on the annulus is a new estimate, not a matter of computing budget.

The certified artifacts also price that ingredient. Across the three positive margins the Region I band populations have similar margin profiles: 11, 3 and 11 bands certify below $3 \cdot 10^{-4}$ at $\kappa = 0.05$, 0.0995 and 0.13, between 55 and 66 below $3 \cdot 10^{-3}$, and 87 to 88 percent of the 1404 bands above $3 \cdot 10^{-2}$. If each band carried a certified bound L_i on the κ-derivative of its certificate value, a band would hold over a κ-radius m_i/L_i of its margin m_i, and re-certifying each band at its own pitch would cover 10^{-2} of margin axis for 1.5 to 1.9 times the cost of one point run at unit derivative bound and 2.5 to 3.8 times at a bound of three, against about $2 \cdot 10^6$ band evaluations for uniform slabs at the pitch of the thinnest band ($\kappa = 0.13$, bound three). The compute is therefore not the obstruction; what is missing is the derivative enclosure itself, one certified κ-derivative per certificate family, with the located branch differentiated implicitly. The constant itself is measurable from the existing artifacts: the band lattice is shared across margins up to the wedge rotation, and 1292 rectangles coincide exactly between the $\kappa = 0.05$ and $\kappa = 0.0995$ manifests, giving certified finite differences of each certificate value along the located branch. The median band moves at $|\Delta S/\Delta\kappa| \approx 7$, and of the ten thinnest shared bands at $\kappa = 0.05$ (margins $2.2 \cdot 10^{-4}$ to $1.6 \cdot 10^{-3}$) nine move at less than 0.5 and four at less than 0.06; the thinnest of them holds over a κ-radius near $2 \cdot 10^{-4}$. Taking the measured difference as L_i on the shared bands and the median on the others gives 3.9 runs per 10^{-2} of margin axis, about 31 Region I run-equivalents for all of $[0.05, 0.13]$. A difference over a 0.05 gap bounds the mean drift rather than the local derivative, so these numbers size the project rather than prove it; the machinery an interval theorem still requires is the certified per-certificate κ-derivative, with the same measurement then repeated for the sweep, union, and star-interior families.

8 Toward the matching lower bound

Krauth and Mézard conjectured the value of the limiting ratio $m_N(\kappa)/N$ for every margin, not only at $\kappa = 0$ [1]; the constant 0.833 is the zero-margin point of the conjectured curve $\alpha_\star(\kappa)$. At $\kappa = 0$ the conjecture is resolved: Ding and Sun [6] proved $\alpha_\star$ is a lower bound with positive probability, the sharp-threshold sequence of Nakajima and Sun [16] upgrades positive probability to convergence in probability, Huang [7] proved the matching upper bound, and the numerical hypotheses of both proofs are certified [9]. Away from $\kappa = 0$ the upper half is

Theorem 4. For the lower half, the replica-symmetric free energy is proved at small constraint density by Talagrand [5] and, for a wider class of activations, by Bolthausen, Nakajima, Sun, and Xu [10]; what has no analytic theorem to certify at present is a lower bound matching $\alpha_\star(\kappa)$. This section delimits that gap: a lemma on why unconditioned moment methods cannot close it at nonnegative margins, an embedded bound that does give nonzero-margin capacity lower bounds on the negative side, and a numerical verification that the κ-dependent inputs of the Ding–Sun program hold at all five certified margins.

8.1 Plain second moments fail at every margin

Write $Z = Z_{N,\kappa}$ for the number of $w \in \{-1,+1\}^N$ satisfying all $m = \lceil \alpha N \rceil$ constraints, and $q_\rho(\kappa) = \mathbb{P}(X \ge \kappa,\, Y \ge \kappa)$ for (X,Y) bivariate standard Gaussian with correlation ρ.

Lemma 6. *For every $\kappa \in \mathbb{R}$ and $\alpha > 0$,*

$$\liminf_{N\to\infty} \frac{1}{N} \log \frac{\mathbb{E}[Z^2]}{(\mathbb{E}Z)^2} \;\ge\; c(\alpha,\kappa) \;>\; 0.$$

Consequently the guarantee of the Paley–Zygmund bound, $\mathbb{P}(Z > 0) \ge (\mathbb{E}Z)^2/\mathbb{E}[Z^2]$, is at most $e^{-cN+o(N)}$, and no bound depending on $\mathbb{E}Z$ and $\mathbb{E}[Z^2]$ alone can certify more, since a distribution placing mass $(\mathbb{E}Z)^2/\mathbb{E}[Z^2]$ at $\mathbb{E}[Z^2]/\mathbb{E}Z$ and the rest at zero matches both moments. For $\kappa \ge 0$ the same conclusion holds for every weighted count $Z_f = \sum_w \prod_a f(\langle g_a, w\rangle/\sqrt{N})$ with $f \ge 0$ supported on $[\kappa,\infty)$ and $0 < \int f\, d\Phi < \infty$.

Proof. $\mathbb{E}Z = 2^N \Psi(\kappa)^m$ by independence over constraints. For a pair w, w' at overlap ρ the constraint fields are bivariate Gaussian with correlation ρ, so counting ordered pairs by Hamming distance k,

$$\mathbb{E}[Z^2] = 2^N \sum_{k=0}^{N} \binom{N}{k} q_{1-2k/N}(\kappa)^m.$$

Fix $\rho_0 \in (0,1)$, let $k^* = \lceil N(1-\rho_0)/2 \rceil$ and $\rho^* = 1 - 2k^*/N$. Keeping the single term k^* and using $\binom{N}{k} \ge e^{NH(k/N)}/(N+1)$ with H the natural-log binary entropy ($H(p) = H(1-p)$, so the exponent may be written with $(1+\rho_0)/2$),

$$\frac{1}{N}\log\frac{\mathbb{E}[Z^2]}{(\mathbb{E}Z)^2} \ge \underbrace{H\Big(\tfrac{1+\rho_0}{2}\Big) - \log 2 + \alpha\big[\log q_{\rho_0}(\kappa) - 2\log\Psi(\kappa)\big]}_{=:\,\psi_{\alpha,\kappa}(\rho_0)} + O(N^{-1}\log N),$$

the error, of order $N^{-1}\log N$, collecting the binomial prefactor, the $|\rho^* - \rho_0| \le 2/N$ continuity of $\rho \mapsto \log q_\rho$ (differentiable, below), and $|m - \alpha N| \le 1$. Now $q_0 = \Psi(\kappa)^2$ and $H(1/2) = \log 2$ give $\psi_{\alpha,\kappa}(0) = 0$. Reflecting the orthant to a joint CDF, $q_\rho(\kappa) = \Phi_2(-\kappa,-\kappa;\rho)$, and Plackett's identity $\partial\Phi_2/\partial\rho = \varphi_2$ with the symmetry $\varphi_2(-\kappa,-\kappa;\rho) = \varphi_2(\kappa,\kappa;\rho)$ give $\partial q_\rho/\partial\rho = \varphi_2(\kappa,\kappa;\rho)$ on $(-1,1)$; at $\rho = 0$ this is $\varphi(\kappa)^2$. Since $\frac{d}{d\rho}H(\frac{1+\rho}{2}) = \frac{1}{2}\log\frac{1-\rho}{1+\rho}$ vanishes at 0,

$$\psi'_{\alpha,\kappa}(0) = \alpha\,\frac{\varphi(\kappa)^2}{\Psi(\kappa)^2} \;>\; 0,$$

so $\psi_{\alpha,\kappa}(\rho_0) \ge \rho_0\,\psi'(0)/2 > 0$ for all small $\rho_0 > 0$; fix such a ρ_0 and set $c = \psi_{\alpha,\kappa}(\rho_0)$. For the weighted counts, the diagonal term of $\mathbb{E}[Z_f^2]$ is $2^N\mathbb{E}[f(X)^2]^m$, so either $\mathbb{E}[Z_f^2] = \infty$ and the

claim is trivial, or $f \in L^2(\Phi)$ and Mehler's expansion gives $\mathbb{E}[f(X)f(Y)] = \sum_{j\geq 0} \rho^j \langle f, \mathrm{He}_j\rangle^2/j!$, whose ρ-derivative at 0 is $(\int x f \varphi\, dx)^2$; for $f \geq 0$ supported on $[\kappa, \infty)$ with $\kappa \geq 0$ the integrand $xf(x)\varphi(x)$ is nonnegative, and it vanishes almost everywhere only if f vanishes almost everywhere off the origin, which $0 < \int f\, d\Phi$ excludes; the argument then runs unchanged. □

The mechanism is the asymmetry: pairs of solutions at small positive overlap are exponentially favored over independent pairs at every margin, which is what the AMP-conditioned second moment of [6] is built to circumvent at $\kappa = 0$, and what any lower-bound proof must circumvent at every other margin. The zero-margin instance of this failure is classical (see the introduction of [6]); the content here is the uniform mechanism and the closure over nonnegative-margin weights. For $\kappa < 0$ the weighted closure genuinely fails, and profitably so: a window weight symmetric about the origin has zero linear Mehler coefficient, which is the next observation.

8.2 A capacity lower bound at negative margins

Proposition 7. *Let $\kappa < 0$ and let $\alpha_c^{\mathrm{sym}}(K) = \log 2/(-\log \mathbb{P}(|X| \leq K))$ denote the symmetric binary perceptron threshold at width K. For every $\alpha < \alpha_c^{\mathrm{sym}}(|\kappa|)$, with probability $1 - o(1)$ the margin-κ perceptron admits a solution, that is, $m_N(\kappa)/N \geq \alpha$.*

Proof. If $|\langle g_a, w\rangle|/\sqrt{N} \leq -\kappa$ for all a then in particular $\langle g_a, w\rangle/\sqrt{N} \geq \kappa$, so the symmetric solution set at width $|\kappa|$ embeds in the margin-κ solution set. The symmetric threshold is α_c^{sym} [11]; for Gaussian disorder the satisfiability side below it holds with high probability by Altschuler's critical-window theorem [14]. (The lognormal limit of Abbe, Li, and Sly [13] is proved for Rademacher disorder, and Perkins and Xu [12] treat the Gaussian case under a single-critical-point hypothesis, so neither alone covers the model here.) □

The bound should be read next to monotonicity. A solution at margin 0 is a solution at every $\kappa < 0$, so $m_N(\kappa) \geq m_N(0)$, and at $\kappa = 0$ the ratio $m_N(0)/N$ converges in probability to $\alpha_\star(0) \approx 0.8330786$ (the lower half by [6] with the sharp-threshold transfer of [16], its numerical hypotheses certified in [9]). Hence for $\kappa < 0$

$$\frac{m_N(\kappa)}{N} \;\geq\; \max\bigl\{\alpha_\star(0),\, \alpha_c^{\mathrm{sym}}(|\kappa|)\bigr\} - o(1) \quad \text{whp.}$$

At the two certified negative margins monotonicity is the stronger input: the embedding gives 0.215 at $\kappa = -0.05$ and 0.655 at $\kappa = -0.45$, against the predicted 0.889 and 1.558. The embedding takes over only for $\kappa < -0.5757$, where $\alpha_c^{\mathrm{sym}}(|\kappa|)$ passes $\alpha_\star(0)$. Li, Schramm, and Zhou [18] prove lower bounds on the satisfiability threshold of the same one-sided perceptron at negative margin by a different route, with the sharpest statements in the $\kappa \to -\infty$ regime.

8.3 The Ding–Sun condition at general margin

The construction underlying the $\kappa = 0$ lower bound of [6] is written at general margin throughout their Section 2: their equations (34)–(48) define, for parameters $(\kappa_{\mathrm{all}}, \alpha_{\mathrm{all}})$, an explicit univariate function $S_{\kappa,\alpha}(\omega)$ on $[\omega_{\min}, 1]$ built from the fixed point $(q_\star, \psi_\star)$ (an entropy term through the two-spin measure $P_{H,D}$, a constraint term through the conditional Mills law φ_ξ, and a convex $\inf_s$ correction), and their Condition 1.2 is the statement $S_{0,\alpha_\star}(\omega) < 0$ for $\omega \notin \{0, 1\}$. The κ-analog of that hypothesis is therefore already defined in their own notation,

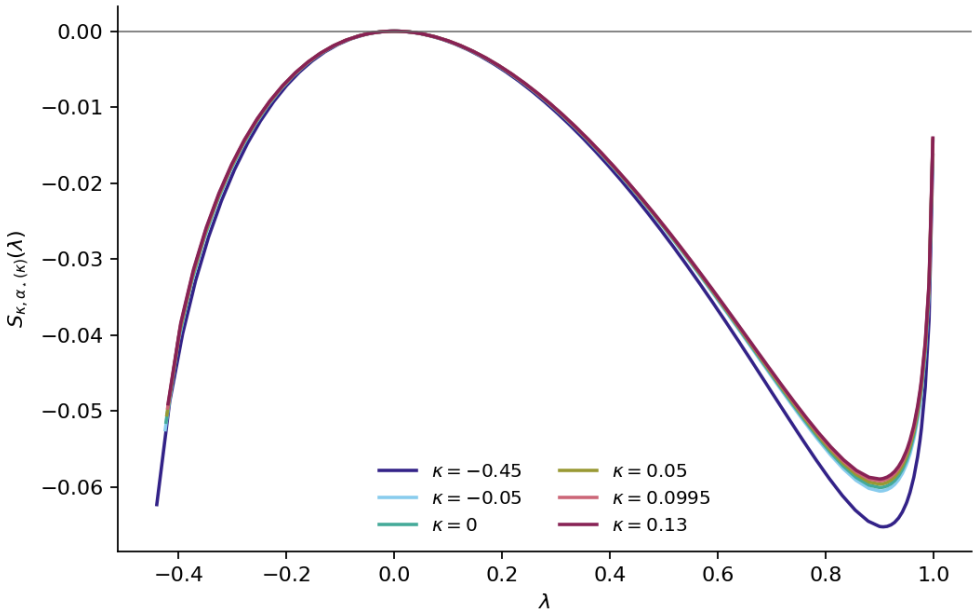


Figure 2: The general-margin Ding–Sun lower-bound condition, evaluated at the five certified margins and $\kappa = 0$: the univariate function $S_{\kappa,\alpha_\star(\kappa)}$ of their equation (48), whose nonpositivity off $\{0, 1\}$ is the numerical hypothesis of their second-moment argument. All six curves are negative off the two roots and nearly coincide.

and every κ-dependent ingredient it consumes is an object this paper certifies: the fixed point, its uniqueness rectangle, and the crossing (Theorem 5 and the deep enclosures of Theorem 4).

We evaluated $S_{\kappa,\alpha_\star(\kappa)}$ in floating point (the driver `ds_condition.py` uses Gauss–Hermite and conditional Gauss–Legendre quadrature) at the five certified margins and at $\kappa = 0$, checking the six endpoint identities $H(0) = P(0) = A(0) = 0$, $H(1) = -H_\star$, $P(1) = -P_\star$, and $G_\star = H_\star + P_\star \approx 0$ (with $H_\star$, $P_\star$ the entropy and constraint halves of the free energy), and that the conditional-mean check resolves at 10^{-12} or better at every margin, as implementation controls. The result is uniform (Figure 2): at every margin the curve is strictly negative off $\{0, 1\}$, at most $-5 \cdot 10^{-5}$ outside windows of half-width 0.02 around the two roots (inside them it climbs to its maxima at the roots, reaching $-1.6 \cdot 10^{-6}$ at the closest grid points), the maximum at $\omega = 0$ is degenerate exactly as at $\kappa = 0$, and the whole family collapses nearly onto a single shape, with $\omega_{\min}$ drifting from -0.441 at $\kappa = -0.45$ to -0.421 at $\kappa = 0.13$. These are diagnostics, not certificates; certifying $S_{\kappa,\alpha_\star} < 0$ off the roots is a univariate problem of exactly the species the Condition 4 machinery already handles, and we expect it to be far easier than the two-variable certification carried out here.

What separates these verified inputs from a general-margin lower bound is the probabilistic body of [6]: the AMP-conditioned moment computation of their Sections 2–7, whose displays carry κ_{all} explicitly; by their Remark 1.5 the assumption $\kappa = 0$ enters only in a few steps, which they indicate. An audit of that body would need, beyond the certified inputs above: a priori bounds at margin (the upper one is free at every κ from the first moment, since $\mathbb{E}Z \to 0$ exponentially above the annealed threshold $-\log 2/\log \Psi(\kappa)$; for $\kappa < 0$ the lower

one is the monotonicity bound above), the state-evolution inputs (generic in the nonlinearity, hence in κ), and the sharp-threshold transfer of [16] if convergence in probability rather than positive probability is wanted. We record this as the concrete route to the exact threshold at the certified margins: the numerical side is done at six points and the analytic side is a κ-generalization audit of an existing proof, not a new construction.

9 Checking the computation

A verification of this size is itself software, and we treat its correctness as part of the result. Five independent layers check the chain, and on the zero-margin verification these layers caught disjoint defect classes.

Ball-arithmetic soundness by construction. Every quantity is an arb ball containing the true value; a PASS is a ball comparison true only when certain. The primitives (monotone endpoint evaluations, mean-value cells, correlated integrands, tail bounds) are shared with the audited zero-margin verification.

Symbolic verification of every identity. The stationarity and $\partial R/\partial q$ identities, the ℓ reformulation (identity, monotonicity, bound), the Nakajima closed form, the general-margin additions to the constraint kernels, and the Bregman collapse are checked in computer algebra; the finite-difference anchors run alongside.

Anchors at $\kappa = 0$. Every pipeline reproduces the certified zero-margin values: the fixed-point rectangle, Huang's AMP and ℓ constants, his C_j and $\det \mathcal{H}$ intervals, and the ported quadrature layer is digit-identical against the zero-margin original at test points.

Exact artifacts and replay. The two-variable chain's certificates carry every sign-critical quantity as outward-rounded decimal packets in canonical JSON, with SHA-256 source and payload binding and atomic no-clobber publication; the stage-2 artifact binds the hash of the Region-I manifest it consumed. Some schedule-geometry fields are ordinary JSON floats; they parameterize display and tiling bookkeeping, not the inequalities. The strip-slab and rectangle files record their enclosures as display strings, and their sign claims are not part of the kernel layer. A documentation check compares the replay instructions with the code, and a fresh-clone test (clone, then run as a reader would) is part of the release. A dependency-free portable checker re-verifies the connection layer from the JSON artifacts alone: the envelope hashes, the stage-2-to-Region-I bindings, the star-interior chain recomputed outward in exact rationals, and the tilings within each serialized chain. It also binds each recorded leaf value to the inequality it stands for (every cell or band marked passing must carry a nonpositive worst-case $\mathcal{S}_\star$ and no subcell failure), so a flipped flag or a mismarked positive value is caught rather than trusted (the transcendental re-evaluation itself remains the ball-arithmetic layer's job). The checker also enforces two global completeness clauses on the band geometry: the innermost ring must close the full circle as a single seam-exact run, and each level's angular support must sit inside the previous level's, which together make every direction's radial chain contiguous out to the radius where its support ends; a deliberately corrupted manifest with all hashes consistently rebound fails exactly this check. The outermost band level is bound to the star radius recorded in the same manifest, which is the reach the stage-2 containment consumes and binds by hash. The Lean tilings carry per-chain scope only; that remains recorded as open hardening.

A second proof system. The exact combinatorial and algebraic layer of the certificate chain (the star-interior inequality chain and the positivity of its perturbation determinant from the

serialized packets, the definiteness of the Loewner majorants, and the gap-free tilings of the sweep grids and the Region-I band schedule) is re-stated over the integers and kernel-checked in Lean 4, following the zero-margin release: per-margin generated files whose kernel-decided theorems cover the 1404-band radius chain and its angular runs, both sweep grid tilings, and the star-interior packet chain, with the Loewner-majorant family at all five margins. The extraction is mirrored in exact rational arithmetic so an extraction bug cannot silently weaken the Lean claims, and a mutation battery (negating one certificate integer per file) rejects every mutant. The slab-level sign claims are not part of this layer: the strip manifests carry display strings rather than exact packets. This layer does not formalize the analytic mathematics; it removes the Python layer from the trusted base for the parts a kernel can decide.

10 What the curve says about learning

For binary weights the Euclidean norm is fixed at $\sqrt{N}$, so a margin constraint is exactly a robustness certificate: if $y_a\langle w, x_a\rangle/\sqrt{N} \geq \kappa$ then the label y_a is invariant under every perturbation of x_a of Euclidean norm below κ. With centrally symmetric i.i.d. designs and independent labels, label folding makes the robust-interpolation event identical to the satisfiability event studied here, so $\alpha_\star(\kappa)$ is the predicted threshold for how many random labels per parameter a one-bit-per-parameter linear classifier can memorize with prescribed robustness radius. The certified $\kappa = 0.05$ bound, for instance, says such a classifier cannot robustly store more than 0.78108 random labels per parameter at that radius, with the guarantee holding against a worst-case adversary in feature space.

Reading the whole curve quantitatively (`results/robustness_price.csv`): on the certified strip the threshold decays log-linearly at rate 1.39 per unit margin, so robust storage capacity halves for every 0.50 of demanded feature-space radius, and the decay continues smoothly beyond (a factor 4.8 from $\kappa = 0$ to 1.5). The efficiency of binary relative to real-valued weights *rises* with the demanded robustness: $\alpha_\star/\alpha_{\text{spherical}}$ grows from 0.417 at $\kappa = 0$ to 0.556 at $\kappa = 1.5$, so one-bit parameters give up proportionally less storage precisely where the robustness requirement is stiffest. On the relaxed side ($\kappa < 0$, tolerating margin violations up to $|\kappa|$) capacity grows toward the annealed bound (the ratio reaches 0.996 by $\kappa = -2$) and the solution space measurably approaches the replica-symmetry-breaking regime (the weak Hessian eigenvalue of the rate function degrades threefold by $\kappa = -1.5$), which is where the replica-symmetric prediction itself should eventually fail. The replica-symmetric cavity picture also predicts the margin distribution *among* solutions: conditionally on the quenched component $\sqrt{q}\,Z$ of a constraint's field, the field on a typical solution is $N(\sqrt{q}\,Z,\, 1-q)$ truncated to $[\kappa, \infty)$, so the gap density across constraints is the mixture $p(h) = \mathbb{E}_Z[\varphi_{1-q}(h - \sqrt{q}\,Z)\,\mathbf{1}\{h \geq \kappa\}/\Psi(u(Z))]$. Computed along the branch at $\kappa = 0.05$ (a float diagnostic, `margin_distribution.py`), the mean gap above the demanded margin compresses from 0.70 at $\alpha = \frac{1}{2}\alpha_\star$ to 0.59 at $\alpha_\star$: solutions are squeezed against the constraint boundary as storage saturates, which is the quantity a margin-type generalization argument would consume, and a natural next target for the certified machinery. These are statements about information-theoretic storage, not about algorithms: for the Ising perceptron the known efficient algorithms stall far below $\alpha_\star$, and the margin curve gives the ceiling against which that algorithmic gap should be measured at every κ. Limits of the reading: nothing here concerns generalization on structured data, correlated features, or hidden layers; κ is a feature-space radius, and translating it to input space requires a Lipschitz bound on the feature map.

The algorithmic gap the curve frames is directly measurable at moderate size. A population annealer (hinge energy on violated margins, geometric schedule matched to the $2/\sqrt{N}$ energy scale; `finite_size_experiment.py`) trained $N = 101$ instances across α at $\kappa \in \{0, 0.05, 0.1\}$. Its success walls (the half-success crossings) fall in the correct κ order and sit below the corresponding $\alpha_\star(\kappa)$ by 0.08–0.12 in α at this budget, while scattered successes persist to the bound itself: at $\kappa = 0.1$ the last appear at $\alpha \approx 0.73$, essentially at $\alpha_\star$, and none beyond. The solutions the annealer does find are atypical in exactly the direction the clustering picture predicts: at low load their mean gap above κ tracks the typical-solution law $p(h)$, while near the wall it holds around 0.65–0.70 as the typical mean descends; the algorithm reaches the wide, over-margined minority. At $N = 201$ under the same per-spin budget the walls sharpen and shift left, so what the annealer traces is a budget-dependent lower envelope of the true algorithmic threshold, whose location and the clustering barrier behind it are studied in [15, 18] on the closely related symmetric and asymmetric binary perceptrons; the found-solution histograms meanwhile track the derived $p(h)$ closely at moderate load and deform toward mid-gaps near the wall. Both effects are schedule-dependent observations, not thresholds; the certified curve is what they are measured against, and the comparison compresses neatly: within each size and budget the wall sits at a κ-independent fraction of $\alpha_\star(\kappa)$ (0.90 at $N = 101$ down to 0.50 at $N = 801$, flat across margins to better than one percent at the largest sizes), so the demanded margin costs the algorithm the same relative capacity it costs the storage bound (Figure 3). At fixed sweeps per spin the fraction falls with N (0.90, 0.75, 0.69, 0.63, 0.50 at $N = 101$, 201, 301, 401, 801), close to a fixed loss of 0.13 per doubling over the measured range, so an early $1/N$ reading of the first three sizes (which would have leveled near 0.59) is refuted by the larger ones; holding a constant fraction of capacity requires superlinearly growing total work. On the budget axis at $N = 201$ the fraction climbs 0.75, 0.79, 0.86 as the budget quadruples twice, but a further quadrupling to 25,600 sweeps per spin gains only 0.02 (0.876): the fixed gain per quadrupling stops at 6,400, and the wall saturates near 0.88 of capacity at this size over the measured range.

The margin price also depends on the weight alphabet. Solving the replica-symmetric system for ternary weights $\{-1, 0, +1\}$ with a self-overlap order parameter (`exploration/ternary_km.py`, anchored by reproducing the binary $\alpha_\star(0) = 0.8330786$ to the certified digits) gives $\alpha_\star^{\mathrm{tern}}(0) = 1.17448$ against an annealed bound of $\log_2 3$, with 1.07951 at $\kappa = 0.05$ and 0.99483 at $\kappa = 0.1$. Per weight *state* the richer alphabet stores more; per weight *bit* it stores less: 0.741 against the binary 0.833 constraints per bit at $\kappa = 0$, and the gap widens monotonically with margin (0.100 at $\kappa = 0.05$, 0.105 at 0.1). At this level of the theory, binary remains the bit-efficient quantization for robust storage. The full eight-point curve (Figure 4, κ up to 0.35) makes this a statement about the whole margin range rather than three samples: per stored bit the binary curve sits above the ternary one everywhere, the absolute gap saturating near 0.11 constraints per bit above $\kappa \approx 0.2$ while the relative advantage grows from 12 to 26 percent, since both capacities fall with margin.

11 Scope and further directions

The machinery is not specific to the binary prior or to this constraint law, and the natural next studies are ordered by how much of it they reuse.

The strip, quantified. The four-piece certificate of Section 6 accepts a κ ball in place of a point, but run on slab hulls with the present evaluator it fails for the reasons measured in

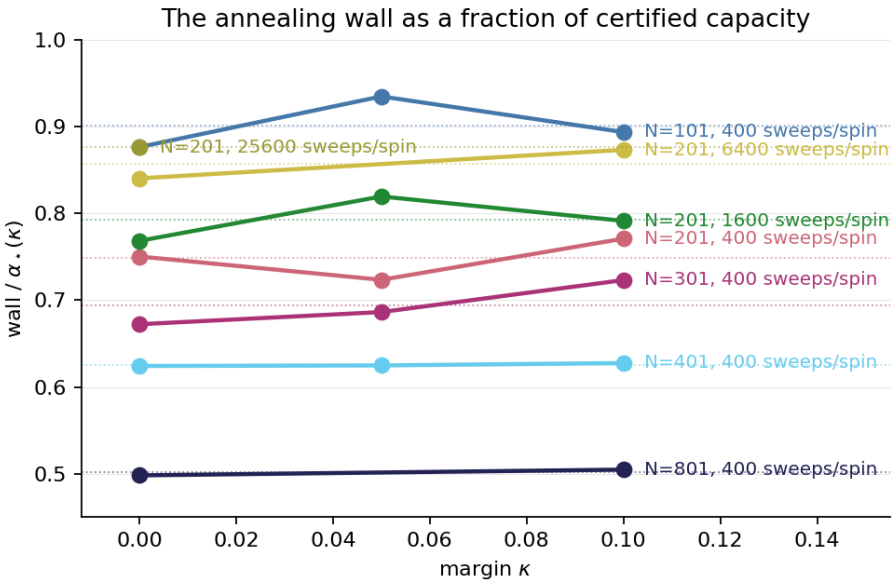


Figure 3: The annealing wall as a fraction of the predicted capacity $\alpha_\star(\kappa)$, across size–budget pairs. Within each run the fraction is flat in κ to a few percent, while its level depends on both size and budget.

Section 7.1. An interval theorem needs two ingredients that are not in this paper: a certified κ-derivative for each certificate family, and quadratic control with explicit remainder bounds on the annulus between the star and the fat far-field cells. Given those, an adaptive κ-subdivision driven by the certified margins would turn Theorem 4 into a statement on the whole strip.

Other margins of the same family. The certified pipeline applies wherever the continuation reaches: positive margins to $\kappa \approx 1.15$ (past which the fixed-tilt center certificate must re-optimize the tilt, as measured in Section 5.1), and negative margins until replica-symmetry breaking territory, where the natural target is the approach (the degradation of the weak eigenvalue and of the ℓ_0 margin) rather than the condition.

Other priors and activations. The moment-coordinate reduction needs only that the profile enter through finitely many moments of a log-concave base pair; replacing $\{-1, +1\}$ by a bounded discrete prior changes Φ and the moment body but not the architecture. The spherical perceptron at margin (Gardner's curve, shown for context in Figure 1) has simpler convexity structure and would serve as a validation target where exact answers are classical.

Teacher–student and structured labels. The planted version replaces the annealed constraint law by a tilted one; the cavity argument and the two-variable function have direct analogues, and the certified-slab design (per-slab rectangles, located boxes, deep bisection) transfers whenever the fixed-point system does.

Toward learning theory. Two quantitative bridges seem within reach of exactly this machinery: certified upper bounds along capacity curves for one-bit compressed classifiers (the κ-robust memorization ceiling above, made uniform over the strip), and certified bounds on the margin distribution achievable at a given $\alpha < \alpha_\star(\kappa)$, which is the object that connects storage geometry to generalization bounds of the classical margin type.

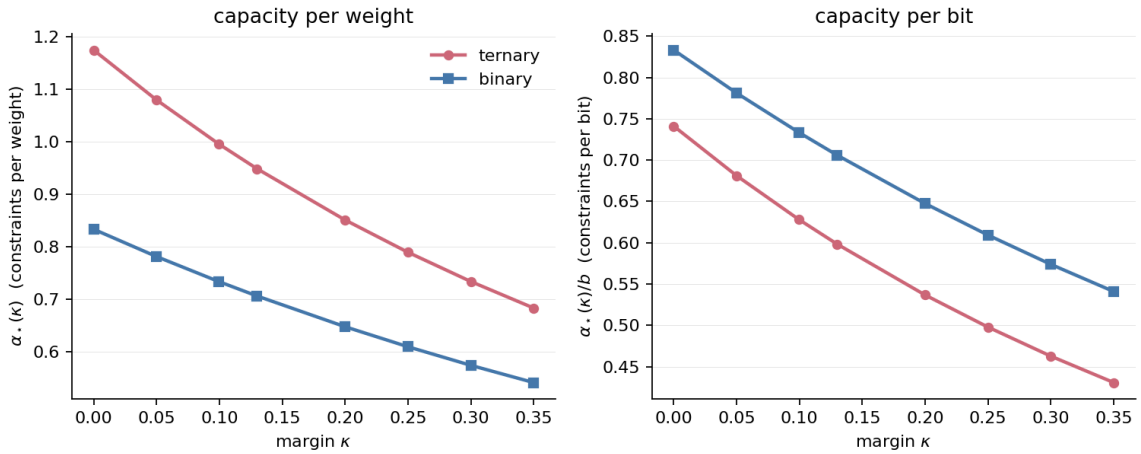


Figure 4: Replica-symmetric capacity of ternary against binary weights, per weight (left) and per stored bit (right, dividing by $\log_2 3$ and 1). The order reverses: the richer prior stores more per weight and less per bit, at every computed margin.

What is proved. The certified statements are: Conditions 1–3 on $\kappa \in [-0.65, 0.10]$, and the same on $(0.10, 0.19]$ with the rectangle clause certified per slab on thin located rectangles (Theorem 5); deep $\alpha_\star$ intervals at five margins; the center Hessian at six; Condition 4 at the margins $\kappa = -0.45$, -0.05, 0, 0.05, 0.0995 and 0.13; and hence Huang's unconditional upper bound at the five nonzero margins (Theorem 4), with the rest of the strip conditional only on Condition 4 at the margin in question. The matching lower bound is open: Ding–Sun's conditioned second-moment reduction is proved at $\kappa = 0$ only, and the embedded bound of Section 8, while rigorous at every $\kappa < 0$, sits far below the replica-symmetric value; nothing here is an exact threshold statement at $\kappa \neq 0$. The floating-point scans are not certificates.

Notation

Each symbol has one meaning throughout. The last column gives the section where the symbol is defined.

Symbol	Meaning	Section
N	number of binary weights	1
w	weight vector in $\{-1, +1\}^N$	1
g_a	a-th pattern, a standard Gaussian vector in $\mathbb{R}^N$	1
m	number of patterns, $m = \lceil \alpha N \rceil$	1
α	pattern density m/N	1
κ	margin	1
φ, Ψ	standard Gaussian density and upper tail	2
E	inverse Mills ratio φ/Ψ	2
Z	standard Gaussian variable	2
$u(z; q, \kappa)$	$(\kappa - \sqrt{q}\, z)/\sqrt{1-q}$	2

Symbol	Meaning	Section
$q,\ \psi$	the two order parameters	2
$P,\ R$	the two halves of the fixed-point map, (1)	2
G	replica-symmetric free energy, (2)	2
$G_\star(\alpha)$	G along the fixed-point branch	2
$\alpha_\star(\kappa)$	the zero of $G_\star$; the predicted capacity	2
(q_0, ψ_0)	the fixed point at $\alpha_\star(\kappa)$	2
$\ell(\hat{z}),\ \ell_0$	the local-concavity function of Condition 3 and its minimum	4.1
$\mu(\hat{z})$	$\mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0}Z))^{-1}]$	4.1
$\tau,\ A,\ b$	$E'(u)$, $1-q_0$ and $\mu(\hat{z}) - A$ in the formula for ℓ	4.1
$\mathcal{H}$	Hessian of Huang's first-moment functional at its maximizer	5.1
η	shifted cavity variable $E\big((\kappa - \hat{H})/\sqrt{1-q_0}\big)/\sqrt{1-q_0}$	5.1
$\dot{H},\ \hat{H}$	independent Gaussians of variances ψ_0 and q_0	5.2
M	$\tanh \dot{H}$	5.2
Λ	a profile, a function of $(\dot{H}, M)$ with values in $[-1, 1]$	5.2
$a = (a_1, a_2)$	the moments $(\mathbb{E}[\dot{H}\Lambda], \mathbb{E}[M\Lambda])$	5.2
K	the moment body, the set of attainable a	5.2
$\mathcal{S}(\Lambda, s)$	Huang's rate function at tilt s	5.2
$\mathcal{S}_\star(\lambda_1, \lambda_2)$	$\inf_{s \geq 0} \mathcal{S}(\tanh(\lambda_1 \dot{H} + \lambda_2 M), s)$	5.2
$\lambda,\ b$	dual points in $\mathbb{R}^2$	5.2
$\Phi(b)$	$\mathbb{E} \log 2\cosh(b_1 \dot{H} + b_2 M)$	5.2
$H(a)$	entropy term as a function of the moments	5.2
$a^\star$	the maximizer $(\psi_0(1-q_0), q_0)$	5.2
s_0	the optimal tilt $\sqrt{1-q_0}$ at $a^\star$	5.2
$\mathcal{F}(a, s)$	$\mathcal{S}(\cdot, s)$ written in moment coordinates	6
$v,\ c,\ t$	ray direction, tilt slope and distance along the ray	6
$\chi_{v,c}(t)$	$\mathcal{F}(a^\star + tv,\ s_0 + ct)$	6
Δ	$(\lambda_1 - 1)\dot{H} + \lambda_2 M$ in the Bregman integrand	6
$V,\ \sigma$	cavity argument and $E(V) - V$	Lemma 3
B	frozen-tilt matrix of a Region I band	7
$m_N(\kappa)$	largest m for which a solution exists	8
$S_{\kappa,\alpha}(\omega)$	Ding and Sun's univariate function on $[\omega_{\min}, 1]$	8

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital.

Declaration of generative AI and AI-assisted technologies

The mathematics, the software, the certificates and the drafting of this paper were done by AI agents, Claude (Anthropic) and Codex (OpenAI), working through the shared memory of the Math Brain project [19, 21, 20]. The author set the objective, supervised the runs, reviewed the mathematics and accepts responsibility for the paper.

Data availability

No data were used. The verification programs, certificates and raw records are available at `https://github.com/yspennstate/ising-perceptron-margin/tree/v2.0`.

The paper is part of the Math Brain project [19].

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